

$$\underline{u} \cdot \underline{v} = |\underline{u}| \cdot |\underline{v}| \cdot \cos \theta \quad 0 \leq \theta \leq \pi$$

$$\underline{u} \cdot \underline{v} = u_i v_i = u_j v_j = \sum_{i=1}^n u_i v_i = u_1 v_1 + u_2 v_2 + u_3 v_3$$

$\downarrow$ dummy

$$\underline{u} = \underline{A} \cdot \underline{v}$$

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$u_i = A_{ij} v_j$$

indice libre

indice dummy

$$u_i = A_{i1} v_1 + A_{i2} v_2 + A_{i3} v_3$$

v_1
+
 v_2
+
 v_3

$$u_1 = A_{11} v_1 + A_{12} v_2 + A_{13} v_3$$

$$u_2 = A_{21} v_1 + A_{22} v_2 + A_{23} v_3$$

$$u_3 = A_{31} v_1 + A_{32} v_2 + A_{33} v_3$$

$$z_i \times (u_i = A_{ij} v_j)$$

$$\alpha = z_i u_i = z_i A_{ij} v_j$$

$$\underline{z} \cdot (\underline{u} = \underline{A} \underline{v})$$

$$\underline{z} \cdot \underline{u} = \underline{z} \cdot (\underline{A} \underline{v})$$

$$\underline{z} \cdot \underline{u} = \underline{z} \cdot \underline{A} \underline{v}$$

$$\alpha = z_j u_j = z_j (A_{jk} v_k) = z_j A_{jk} v_k = z_i A_{ij} v_j$$

$$\alpha = \underline{z} \cdot \underline{u}$$

$$u_i = \sum_{j=1}^n A_{ij} v_j$$

$$\alpha = \sum_{i=1}^n z_i u_i = \sum_{i=1}^n z_i \left(\sum_{j=1}^n A_{ij} v_j \right) = \sum_{i=1}^n \sum_{j=1}^n z_i A_{ij} v_j$$

$$n_i=1 \quad z_1 \cdot (A_{11} v_1 + A_{12} v_2 + A_{13} v_3)$$

$$z_1 \cdot A_{11} v_1 + z_1 A_{12} v_2 + z_1 A_{13} v_3$$



a) $\delta_{ii} = 3$

Autre problème 3D ($n=3$)

$$\delta_{ii} = \delta_{11} + \delta_{22} + \delta_{33} = 3$$

$$\text{Si } n=2$$

$$\delta_{ii} = \delta_{11} + \delta_{22} = 2$$

tensor de 2^{de} ordre
équivalente à tensor identité

$$\delta_{ij} =$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$i=j \Rightarrow \delta_{ij} = 1$$

$$i \neq j \Rightarrow \delta_{ij} = 0$$

$$tr(I)$$

Range 0 $\rightarrow$ escalar

b) $\delta_{ij} \delta_{ij} = \delta_{ii} = \delta_{jj} = 3$
 $\rightarrow$ no opete solo $i=j$

$$\delta_{ij} v_j = v_i$$

$$\delta_{ij} A_{jk} = A_{ik}$$

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & \dots \\ \vdots & \vdots & \vdots \end{bmatrix}$$

9 ecuaciones

$A_{zz} B_{gg} + A_{jj} B_{kk} +$
 ~~$A_{kk} B_{kk}$~~
 $A_{ij} B_{ij}$
 ~~$A_{ii} B_{ii}$~~
 $A_{km} B_{km}$

$A_{ij} = C_{ijkm} u_k C_m$
 Tensor 2do orden 4o 1o 1o

$A_{ij} - C_{ijkm} u_k C_m = 0$

son libres

$A_{ij} - C_{ijet} u_e C_t = 0$

$\rightarrow C_{ijrs} u_r C_s$

$A_{pg} = C_{pgkm} u_k C_m$

$A_{ij} = C_{ijkm} u_k C_m = \rightarrow 2D$

$B_{ij} = C_{ij1m} u_1 C_m + C_{ij2m} u_2 C_m =$

$A_{ij} = C_{ij11} u_1 C_1 + C_{ij12} u_1 C_2 + C_{ij21} u_2 C_1 + C_{ij22} u_2 C_2 =$

$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$
 4 ecuaciones

$A_{11} = C_{1111} u_1 C_1 + C_{1112} u_1 C_2 + C_{1121} u_2 C_1 + C_{1122} u_2 C_2$

$A_{22} = C_{2211} u_1 C_1 + \dots$

$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 \end{pmatrix} + 1$

c) Simbolos de permutacion

e_{ijk}

$\rightarrow$ rango 3

e_{ijk}	$\overrightarrow{123}$	$\overrightarrow{231}$	$\overrightarrow{312}$	$+1$
	$\overrightarrow{213}$	$\overrightarrow{132}$	$\overrightarrow{321}$	-1
	112	221	121	111

$e_{123} = 1$

$e_{213} = -1$

$e_{112} = 0$

$e_{211} = 0$

$e_{231} = 0$

$$e_{ijk} = -e_{kji}$$

$$\overbrace{e_{213}}^{-1} = - \overbrace{e_{123}}^{+1}$$

-1 = -1



$$e_{ijk} = -e_{jik}$$

$$e_{112} = -e_{121} \Rightarrow 0 = -0 = 0$$

$$e_{ijk} = e_{jki} = e_{kij}$$

$$A_{jksl} = e_{ijk} e_{isl} = \delta_{js} \delta_{kl} - \delta_{ji} \delta_{ks}$$

$$A_{jkkj}$$

tensor de orden = escalar

$$j=k \Rightarrow \delta_{jk} = 1$$

$$j \neq k \Rightarrow \delta_{jk} = 0$$

$$3D) \Rightarrow e_{ijk} e_{jki} = e_{ijk} e_{ijk} = \delta_{jj} \delta_{kk} - \delta_{jk} \delta_{kj} = 3 \cdot 3 - 3 = 6$$

$\delta_{jj} = 3$

$$A = A^T$$

$$\delta_{jk} = \delta_{kj}$$

$$A_{ij} = (A_{ji})^T = A_{ji}$$

$$\alpha \cdot \rho = \rho \cdot \alpha$$

$$e) e_{ijk} A_j A_k = 0$$

$$e_{ijk} A_j A_k = \frac{1}{2} (e_{ijk} A_j A_k + e_{ijk} A_j A_k) =$$

$$= \frac{1}{2} (e_{ijk} A_j A_k - e_{ikj} A_k A_j) =$$

$$= \frac{1}{2} (e_{ijk} A_j A_k - e_{ijk} A_j A_k) = 0$$

$$= \frac{1}{2} (e_{ipq} A_p A_q - e_{ipq} A_p A_q)$$

$p \rightarrow j \ q \rightarrow k$ $p \rightarrow k \ q \rightarrow j$

$$\frac{1}{2} (e_{ijk} A_j A_k + e_{ijk} A_j A_k) =$$

$$\frac{1}{2} (e_{ijk} A_j A_k + e_{ipq} A_p A_q) =$$

$$\frac{1}{2} (e_{ijk} A_j A_k + e_{ipq} A_q A_p)$$

$$\overbrace{e_{ijk} A_j A_k}^{a_i} = - \overbrace{e_{ikj} A_k A_j}^{a_i} = - \overbrace{e_{ijk} A_j A_k}^{a_i}$$

rearrange dummy

$$\left. \begin{array}{l} a_1 = 0 \\ a_2 = 0 \\ a_3 = 0 \end{array} \right\} \left. \begin{array}{l} a_i = -a_i \Rightarrow a_i = 0 \end{array} \right\}$$

$$e) \delta_{ij} \delta_{jk} = \delta_{ik}$$

→ not 0 si i=j
 $\delta_{ij} A_{jk} = A_{ik}$

$$f/k = \delta_{ij} e_{ijk} = e_{iik} \xrightarrow{e_{111} + e_{221} + e_{331} = 0} = 0$$